Mathematics and Further Mathematics – the physicist's toolbox

Before the physics courses proper, everyone had to take the preliminary courses Mathematics and Further Mathematics. They supplied the tools: analysis, linear algebra, complex functions, vector calculus, differential equations, integral equations, and Laplace and Fourier transforms. Almost everything in the rest of the physics degree is built on them. The page has interactive figures on Taylor series, matrices and eigenvectors, the Fourier transform and the Laplace plane.

> University of Sussex, Mathematics and Further Mathematics (Preliminary Course, 1975–78): As I recall, the courses covered mathematical analysis, linear algebra, complex variable analysis, vector calculus, differential equations, integral equations, and Laplace and Fourier transforms. Three of the topics were taken further in final courses and have pages of their own: complex variable theory, Calculus III and applied differential equations. The page describes the topics as they are normally taught; it is not a reproduction of the original courses.

Mathematical analysis

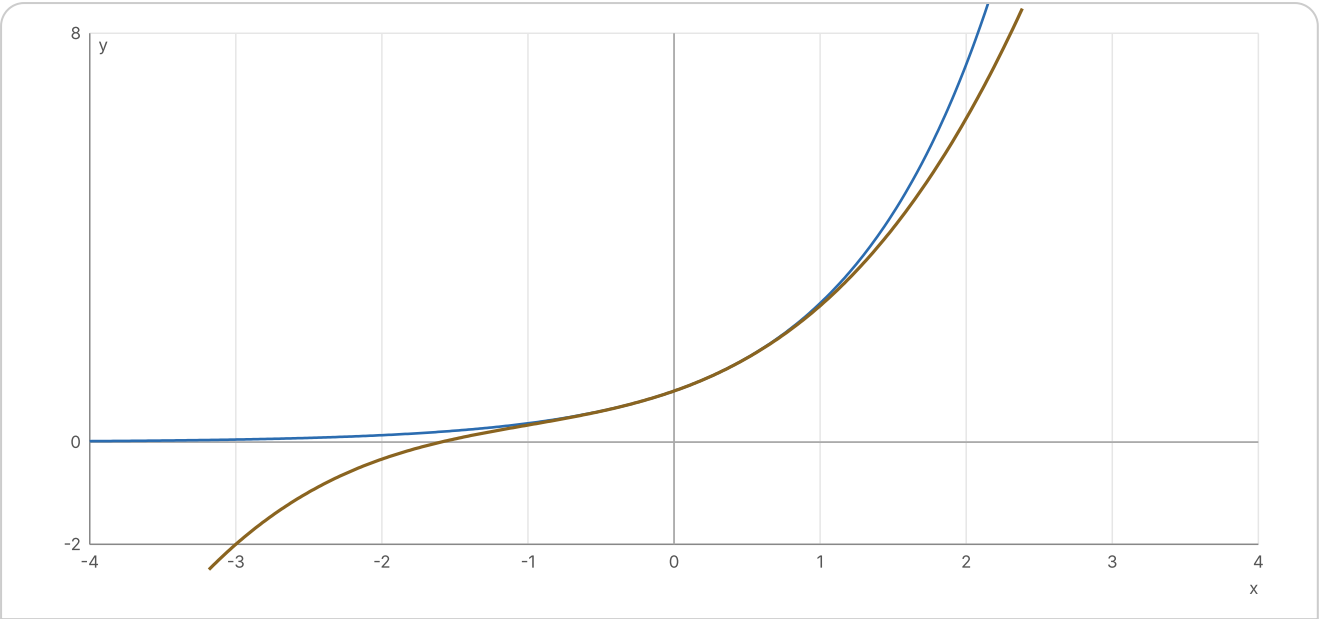
Newton and Leibniz invented the differential and integral calculus in the late seventeenth century, but they worked with "infinitely small" quantities without being able to say exactly what they were. Bishop Berkeley mocked them as "the ghosts of departed quantities". Only in the nineteenth century did Cauchy, Bolzano and Weierstrass give analysis a firm foundation with limits: $\lim_{x\to a} f(x) = L$ means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$. With that definition continuity, differentiability, integrals and the convergence of series can all be made precise.

For the physicist perhaps the most important result is the mean value theorem and its successor, Taylor's theorem: near a point a well-behaved function can be approximated by a polynomial, and the error can be estimated. That is why so many laws of physics are linear – Hooke's law, Ohm's law, the harmonic oscillator – they are the first term of a Taylor expansion.

Taylor series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)\,x + \frac{f''(0)}{2} x^2 + \cdots$$

A power series converges within a *radius of convergence* R and diverges outside it. For e^x and $\sin x$, $R = \infty$; for $\ln(1+x)$, $R = 1$, because the function goes to $-\infty$ at $x = -1$. But $1/(1+x^2)$ is well behaved for all real x – and still $R = 1$. The explanation lies in the complex plane: the function has poles at $x = \pm i$, at distance 1 from the origin (see complex variable theory).

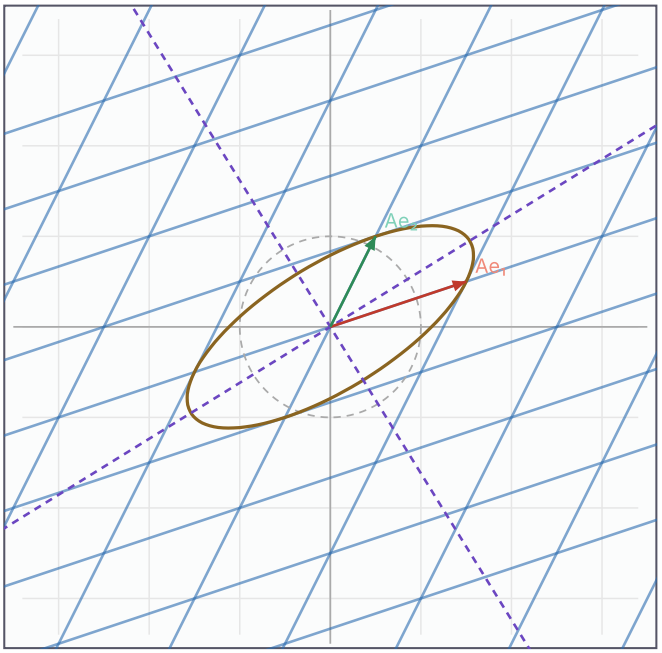


Linear algebra

Linear algebra is about vectors and linear maps – and so about systems of equations, changes of coordinates and symmetries. An $n \times n$ matrix A maps vectors to vectors, and the determinant $\det A$ is the factor by which volumes are scaled. If $\det A = 0$, space collapses, and the system $A\vec{x} = \vec{b}$ has either no solutions or infinitely many.

The most important ideas for physics are *eigenvalues and eigenvectors*: $A\vec{v} = \lambda\vec{v}$. An eigenvector is only stretched, not turned. The eigenvalues are the roots of the characteristic polynomial $\det(A - \lambda I) = 0$. Symmetric (and Hermitian) matrices have real eigenvalues and perpendicular eigenvectors – which is why coupled oscillations have well-defined normal modes (see coupled pendulums), a rigid body has three principal axes of inertia, and in quantum mechanics the results of measurements are the eigenvalues of Hermitian operators. Werner Heisenberg's "matrix mechanics" of 1925 was linear algebra before he himself knew it; it was Max Born who recognised his tables as matrices.

Matrices in pictures

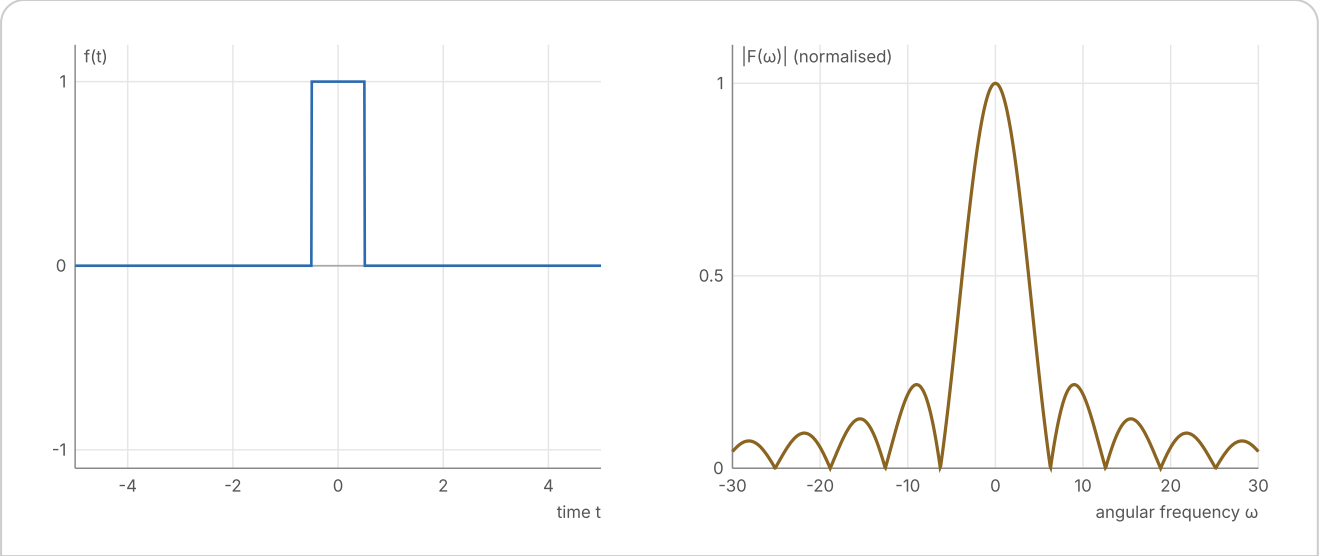


The Fourier transform

In 1807 Joseph Fourier claimed that any periodic function can be written as a sum of sine and cosine oscillations (see Fourier and heat conduction on the page on differential equations). Let the period go to infinity, and the series becomes an integral, the Fourier transform:

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt, \qquad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)\, e^{i\omega t}\, d\omega$$

It translates between time and frequency (or position and wavenumber) and has a number of useful properties: differentiation becomes multiplication by $i\omega$, and a convolution becomes a product. That makes it indispensable in optics, where a lens performs the Fourier transform of the light (see Fourier optics), in quantum mechanics, where the position and momentum wave functions are each other's Fourier transforms, and in acoustics and signal processing (see signal analysis).

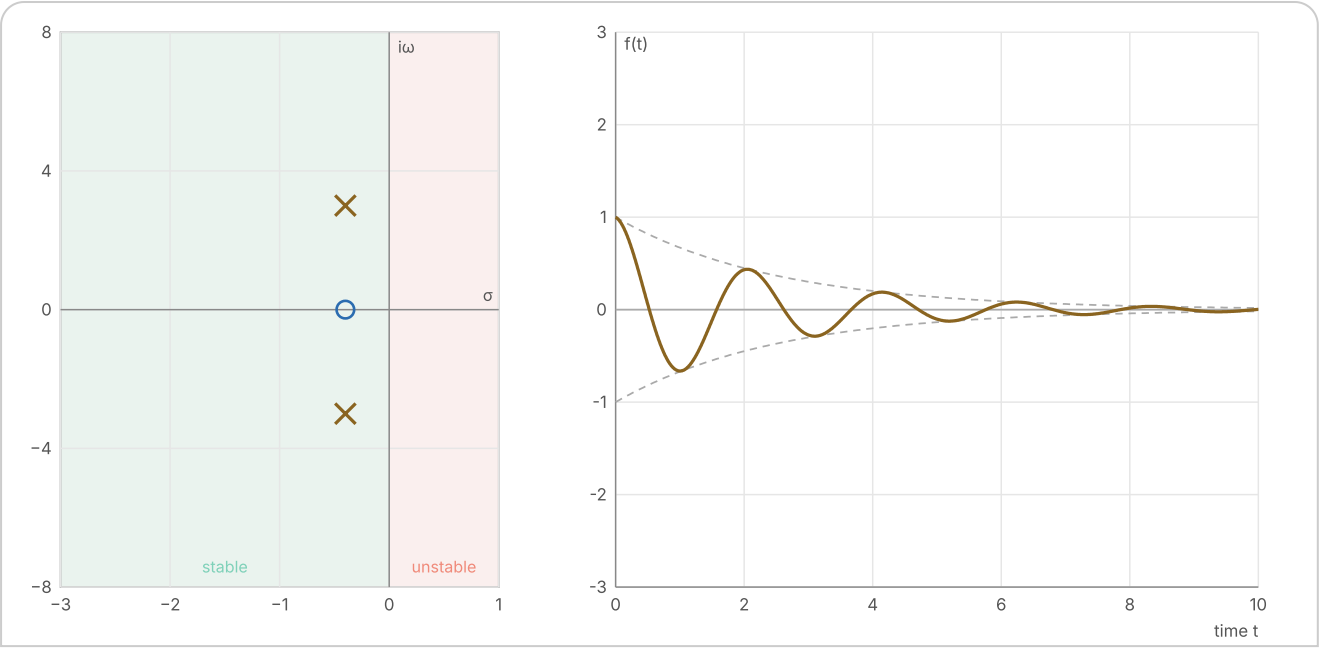


The Laplace transform

The Laplace transform is the Fourier transform's sister, adapted to problems that start at time $t = 0$:

$$F(s) = \mathcal{L}\{f\}(s) = \int_0^{\infty} f(t)\, e^{-st}\, dt, \qquad s = \sigma + i\omega$$

The derivative becomes $\mathcal{L}\{f'\} = sF(s) - f(0)$, so a linear differential equation with initial conditions becomes an algebraic equation in s. One solves it, splits it into partial fractions and looks the result up in a table. It was Oliver Heaviside who in the 1880s began calculating like this with the operator $p = d/dt$ – without proof, to the despair of mathematicians; the Laplace transform later gave the method a sound foundation (see Laplace). In electrical and control engineering the transfer function $H(s)$ is the standard language: its *poles* – the values of s where $H(s)$ becomes infinite – determine the system's oscillations and damping (see circuit theory).



Integral equations

In an integral equation the unknown function appears under an integral sign. The two classic types are named after Vito Volterra and Ivar Fredholm:

$$\int_a^x K(x,t)\,y(t)\,dt = f(x) \quad \text{(Volterra)}, \qquad y(x) = f(x) + \lambda \int_a^b K(x,t)\,y(t)\,dt \quad \text{(Fredh}$$

The first integral equation was solved by Niels Henrik Abel in 1823: down what curve must a ball slide for the time of descent to depend in a given way on the starting height? For a constant time the answer is the cycloid – the tautochrone curve Huygens used for his pendulum clock. If the kernel has the form $K(x-t)$, the integral is a convolution, and the Laplace transform makes the equation algebraic. Fredholm's theory of 1903 led directly to Hilbert's infinite-dimensional spaces – the mathematical home of quantum mechanics.

Many differential equations with boundary conditions can be rewritten as integral equations with a *Green's function* as kernel. In acoustics the Kirchhoff–Helmholtz integral gives the sound field everywhere from the pressure and velocity on a surface – the basis of the boundary element method (BEM), now used to calculate sound radiation from machines and loudspeakers (see machinery noise). In quantum scattering theory the Lippmann–Schwinger equation is an integral equation for the wave function.

The other topics

- **Complex functions** – analytic functions, Cauchy's theorem and residues: see complex variable theory.

- **Vector calculus** – gradient, divergence, curl and the theorems of Gauss and Stokes: see Calculus III.
- **Differential equations** – oscillations, the phase plane and partial differential equations: see applied differential equations.
- **Numerical methods** – when the equations cannot be solved exactly: see numerical mathematics.

Since 1978

- **FFT everywhere:** Cooley and Tukey's fast Fourier transform of 1965 is now in every mobile phone, in MP3 and JPEG and in all wireless communication (OFDM in wifi and 4G/5G).
- **Wavelets** (Morlet, Grossmann, Daubechies in the 1980s) give resolution in both time and frequency and are used in JPEG 2000 and in the analysis of gravitational-wave signals.
- **Linear algebra at scale:** Google's PageRank (1998) is an eigenvector of a matrix with billions of rows, and neural networks are at bottom chains of matrix multiplications.
- **Computer algebra** (Maple 1982, Mathematica 1988) now does integrals, Laplace transforms and eigenvalue problems that once needed books of tables.

Further reading

- Mary L. Boas: *Mathematical Methods in the Physical Sciences* (3rd ed., Wiley 2006)
- Erwin Kreyszig: *Advanced Engineering Mathematics* (10th ed., Wiley 2011)
- Gilbert Strang: *Introduction to Linear Algebra* (6th ed., Wellesley-Cambridge 2023)
- Ronald N. Bracewell: *The Fourier Transform and Its Applications* (3rd ed., McGraw-Hill 2000)